

Multiway Analysis of Bridge Structural Types in the National Bridge Inventory (NBI)

A Tensor Decomposition Approach

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Abstract— The National Bridge Inventory (NBI) has detailed information on over 600,000 bridges nationwide. The data, which spans a period of more than 20 years can be very useful for analyzing and modeling bridge performance. Previous analysis methods employ a 2-dimensional view of data which may result in the loss of subtle trends and changes in the data. Bridge data is inherently multidimensional and as such there is an added advantage in analyzing it in its natural multidimensional state. This paper focuses on the use of a multiway data analysis approach known as tensor decomposition to analyze the structural deficiency rate with respect to states, bridge structural types and time. The tensor decomposition approach is able to reveal clusters and patterns which are not easily perceptible when using conventional statistical tools.

Keywords—Tensors; Multiway Data; Bridges; Data Analysis

I. INTRODUCTION

Bridges are critical infrastructure whose safe functioning is required for economic activities to proceed unperturbed. They form indispensable links to various locations nationwide for commuters and freight daily. As stated by the American Society of Civil Engineers (ASCE) in the 2013 report card for America's Infrastructure, the health of the nation's bridges are tied to its ability to compete in the global marketplace [2] which makes bridge asset management a top priority of Federal, state and local government agencies. To efficiently manage and maintain bridges, data is collected regularly by bridge operators, owners and maintenance engineers. The National Bridge Inventory (NBI) is the largest collection of bridge data in the world [3] and is owned and maintained by the Federal Highway Administration (FHWA). The NBI has detailed information on over 600,000 bridges which meet specified criteria in the United States. Information on bridges includes geographical location, daily traffic, design load, construction types among others. Bridges in the inventory can be classified as structurally deficient or functionally obsolete. A bridge is rated as structurally deficient if load carrying members are in a poor condition or significantly below design standards and or if a waterway frequently overtops the bridge during floods [4]. A functionally obsolete bridge simply refers to one whose design is outdated. This may be the result of increased traffic or the absence of shoulders. Thus, a functionally obsolete bridge may be structurally sound but may

not have adequate clearance for oversized vehicles or can be the cause traffic jams [9].

Almost 11% of the bridges in the NBI database are structurally deficient. The FHWA estimates the cost of repairing all the deficient bridges will be about \$76 billion [10]. With declining gas tax revenues and other federal and state budgetary constraints, raising this amount to solve the problem at once may not be economically sustainable. This implies that agencies involved in the maintenance of bridge assets employ strategic management practices that will ensure efficient use of resources. The basis of a strategic bridge management approach is in-depth knowledge on bridge deterioration which is gained through data analysis and performance modeling of bridge assets.

However, exploratory data analysis of NBI data with conventional statistical techniques becomes a challenge due to the size and complexity of data sets. Bridge data is inherently multidimensional. In order to leverage the data to fully understand bridge condition and behavioral trends, it is essential to analyze bridge data in its natural multidimensional state since traditional 2D analysis may result in loss of information. In this paper, tensor factorization, a multiway data analysis tool is used to analyze 22 years of bridge data from the National Bridge Inventory. Tensors are multiway arrays and tensor decomposition simply refers to reducing data into lower order forms which can be used in clustering, prediction and exploration. The focus of this paper will be on the use of tensor decomposition to analyze distribution of structurally deficient bridges with respect to structural types across Washington DC, Puerto Rico and the 50 states. The bridge types under consideration include slab, box beam, tee beam, stringer/multibeam girders, arch deck and girder & floorbeam systems.

This paper has five main sections. The first section is the introduction followed by the literature review which covers previous research in bridge data analysis as well as a brief definition of the basic concepts of tensor factorization. The third section is the methodology which details the approach used in analysis. The results obtained from analysis are discussed in the fourth section and the conclusion and recommendations in the fifth and final section.

II. LITERATURE REVIEW

The Federal Highway Administration (FHWA) is responsible for keeping the nation's highway system which includes over 600,000 bridges in safe operation. The FHWA works in conjunction with the states to achieve this aim. The U.S. Congress and President established uniform standards for bridge inspection known as the National Bridge Inspection Standards (NBIS) after the 1967 collapse of the Silver Bridge between Ohio and West Virginia [8]. The NBIS requires biennial safety inspections for bridges longer than 20 feet on public roads. The information gathered from regular bridge inspections are maintained for future use by agencies in the National Bridge Inventory. Despite having over 20 years of detailed information on bridges, bridge asset management is still largely based on expert opinions and assumptions. There is therefore the need to model bridge performance over time in order to prevent failure which can result in negative impacts on the economy and the social life of users.

In past, NBI data has been analyzed using conventional statistical 2D techniques. Nien-Chu [3] performed exploratory data analysis of NBI data. In the work, the spatial and temporal patterns of bridge performance with respect to various factors were analyzed. The analysis also used Geographical Information Systems (GIS) tools to provide geolocation information which is very useful for policy-makers. Cruz M. [11] also evaluated the factors that affect deterioration and conceptualized a Markovian model for prediction.

A tensor is defined as a multidimensional array. This concept of tensors is different from tensor fields in mathematics and physics which refer to concepts such as stress tensors. Tensor decompositions were started by Hitchcock in 1927 and Cattell in 1944. However, the area gained much recognition in the 1960's as a result of work by Tucker. Tensors have been applied in psychometrics, chemometrics, data mining, data exploration, graph analysis, neuroscience and many other fields to date [5]. Conventional data analysis involves reordering data as a matrix and analyzing it. However, some parts of the information hidden in the data's natural structure may be lost. As a result, analyzing data in its multidimensional state is advantageous and tensors or multiway data analysis provide a way to achieve it.

The dimensions of a data array is referred to as modes or ways. Fig. 1 below shows a 3-way array with modes (dimensions) i , j and k . There are two main methods by which multidimensional data are decomposed. They are Canonical Decomposition Parallel Factors (CANDECOMP/PARAFAC)

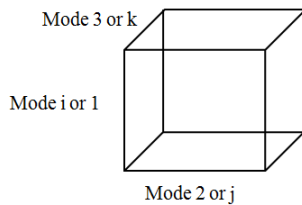


Fig. 1. 3-way array

and Tucker decomposition. The PARAFAC model decomposes the 3-way data array $\mathbf{X}$ such that:

$$x_{ijk} = \sum_{f=1}^{NF} a_{if} b_{jf} c_{kf} + e_{ijk} \quad (1)$$

Where a_{if} , b_{jf} and c_{kf} are the elements of the three loading matrices A ($n_i \times NF$), B ($n_j \times NF$) and C ($n_k \times NF$). NF refers to the number of factors required to approximate the data tensor and e_{ijk} is the error term. The PARAFAC model is a constrained one having the same number of modes for each loading matrix. A core consistency criterion known as core consistency diagnostic (CORCONDIA) is used to determine the level of complexity of the model chosen.

The Tucker model decomposes data such that:

$$x_{ijk} = \sum_{p=1}^P \sum_{q=1}^Q \sum_{r=1}^R a_{ip} b_{jq} c_{kr} g_{pqr} + e_{ijk} \quad (2)$$

a_{ip} , b_{jq} and c_{kr} are the elements of loading matrices A ($n_i \times P$), B ($n_j \times Q$) and C ($n_k \times R$). The core array is denoted as g_{pqr} , and e_{ijk} is the error term. The squared element of the core array reflects the interactions among the three modes. A major advantage of this model is the flexibility of the choice of the number of components in each mode [6].

Tensor factorization is a promising and relatively new way of analyzing large complex multidimensional data. Recent applications include but are not limited to soil pollution analysis [6], fluorescence spectroscopy, bioinformatics, signal processing and computer vision [7]. Modelling bridge data using tensor factorization has immense potential to reveal latent information and patterns about bridge behavior that is not easily perceived with conventional statistical approaches.

III. METHODOLOGY

A. Data

Data used for multiway analysis of bridge structures was obtained from the tables of frequently requested NBI information from NBI database. The data was on structurally deficient and functionally obsolete bridge structure types by state (<http://www.fhwa.dot.gov/bridge/struct.cfm>. Accessed: August 18, 2014). All 22 years of data from 1992 to 2013 made available by NBI was used in the analysis. The data comprised total number of various bridge structural types including slab, tunnel, culvert, frame, multiple girders, orthotropic, suspension among others across DC, Puerto Rico and the 50 states.

For each year, there were 8 tables; (i) total number of various bridge structural types for all states (ii) total number of structurally deficient bridge types (iii) total number of functionally obsolete bridge types (iv) total number of deficient bridges (v) total area of bridge various bridge structural types across all states in square meters (vi) total area of structurally deficient bridges for states (vii) total area of functionally obsolete bridge structural types in various states and (viii) total area of deficient bridges.

B. Data Preprocessing

Preprocessing of bridge data for tensor analysis was performed in 3 main steps. The first step is data selection which involves the selection of relevant parts of the data for

analysis. After data selection, the structural deficiency (SD) rate which is discussed in more detail in this section is calculated in the second step. Last but not least, the final step involves formation of the bridge data tensor used in the analysis.

In data selection, only tables regarding counts of structurally deficient bridges and total counts of bridges were considered. Bridge structural types present in all 50 states and also District of Columbia (DC) and Puerto Rico (PR) were selected. Bridge structural types that were absent in some states were eliminated from the analysis. The 6 bridge types selected were: (i) slab (ii) Stringer/Multibeam or Girder (iii) Girder & Floorbeam system (iv) Tee Beam (v) Box beam or Girders (Multiple) and (vi) Arch-Deck. This was done to ensure an objective comparison of bridge structural types across the entire nation.

The SD rate is calculated for the 6 structural types over all the states. The SD rate is defined as the proportion of bridges of a specific type which are structurally deficient. It is expressed as:

$$SD_{i,j} = \frac{n_{i,j}}{\sum_i n_{i,j}} \quad (3)$$

Where $SD_{i,j}$ is the structural deficiency rate of bridges of structural type i in state j ;

$n_{i,j}$ is the number of bridges of structural type i in state j ; and $\sum_i n_{i,j}$ is the total number of all bridge structural types in state j . In the final step, the data tensor or multiway array is formed in the MATLAB environment. It is a 3-way tensor with 50 states and DC and PR in one mode, 6 structural types in the second mode and 22 years in the third mode. The data tensor was of dimensions $52 \times 6 \times 22$ with no missing entries. See Fig. 2 below for an illustration of the multiway data set created for the analysis. The data tensor created is then analyzed using the MATLAB N-way Toolbox [1].

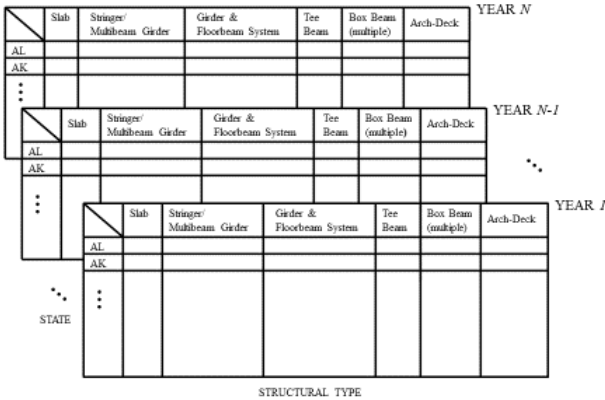


Fig. 2. Three-way dataset for multiway analysis of bridge structural types

IV. RESULTS

Descriptive statistics of the data was performed. Fig. 3 shows the median SD rate of various bridge structural types nationwide over the 22 year analysis period. By inspection, the girder and floorbeam system indicated by the blue color had consistently very high SD rates in most states. The box beam or girders (multiple) and slabs had lower SD rates compared to other structure types. In Fig. 4, Oklahoma, Mississippi, Pennsylvania and Rhode Island had relatively higher mean SD rates for all the bridge types especially girder and floorbeam systems. The mean and median SD rates in Mississippi were 0.814 and 0.809 respectively which means that on average about 80% of girder and floorbeam systems in Mississippi are structurally deficient annually. Florida had the lowest SD rates for all bridge structure types under consideration.

Box plots were used to show distributions of SD rate among various structural types. See Fig. 5. The box plots show the mean SD rates for structural types across all the states over the 22-year analysis period. These were obtained by averaging the mean SD rates for structure types in all the states under consideration.

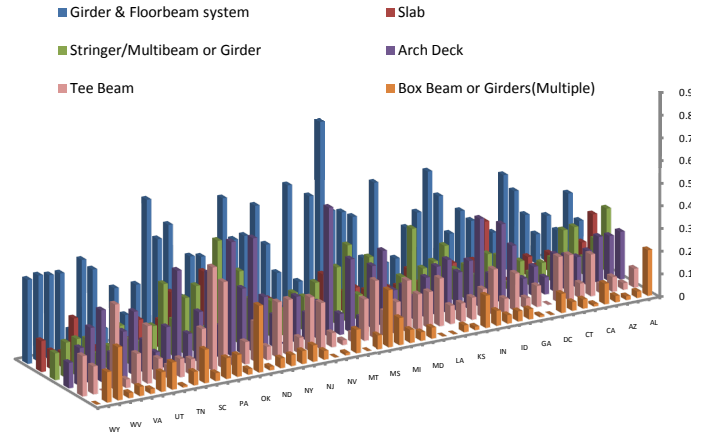


Fig. 3. 3D bar plot of median SD rates for structural types across the US

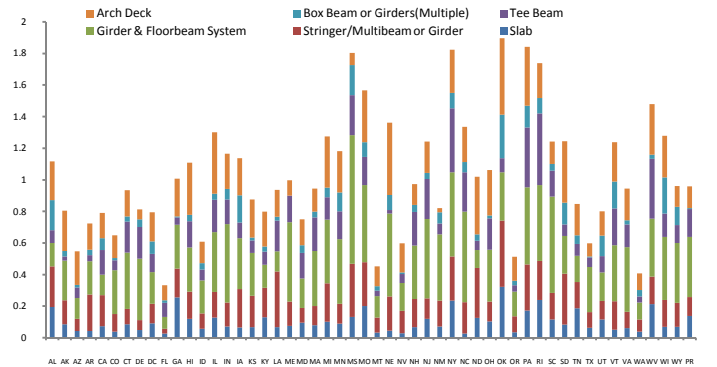


Fig. 4. Mean SD rates for various bridge types

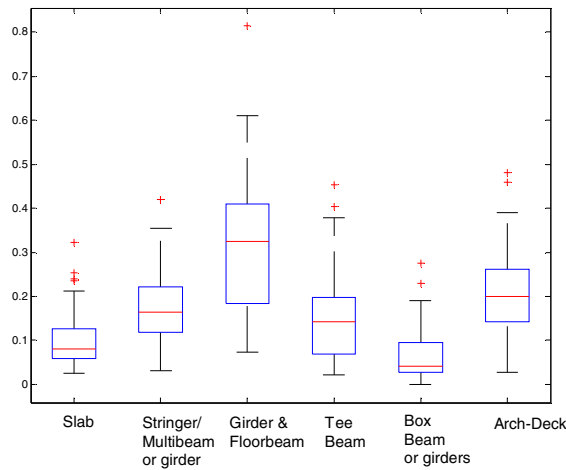


Fig. 5. Box plots for mean SD rates for structural types over 22-year analysis period

The median SD rate for girder & floorbeam construction in Fig. 5 was the highest with a value of 0.3255. Arch-Deck, Stringer/Multibeam or girder and Tee Beams had median SD rates over all states within the 22-year analysis period of 0.1995, 0.1651 and 0.1434 respectively. The slabs had the lowest median SD rate of 0.0813. The box plots which gave a clear indication of trends in structural type deterioration imposed a 2-dimensional form on the data by averaging across states and years. Fig 6 also shows the average trend in SD rates across all states. There is a general recognizable downward trend in the SD rate in the country which indicates improved maintenance practices over the years. It can be observed from the descriptive statistics of the data that there is always some form of averaging or fusion of data points which can result in the loss of valuable information and can lead to misleading conclusions about the data.

Multiway data analysis is carried out using the MATLAB N-way Toolbox [1]. Several PARAFAC models were fitted to the data. The 2-way unconstrained PARAFAC model was the most appropriate model capturing 87.62% of the variance with a core consistency of 92.016% which implies a very stable

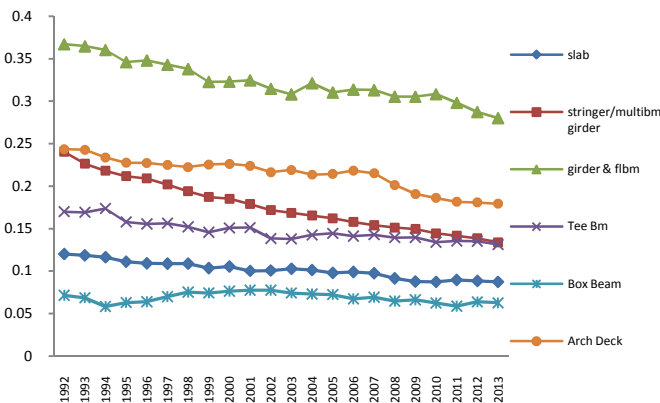


Fig. 6. Mean SD trend rate across all states

model. Table I shows the various models that were fitted to the data.

Model number 4 was selected because a higher percentage variance of 87.62% was captured with a corresponding high level of core consistency of 92.01%. The number of iterations was 134. A core consistency of 70% and above is ideal since such models are much more stable. The 2-component unconstrained PARAFAC model yielded factor matrices for states, structure type and years. Plots of factors revealed information about the bridge data.

Fig. 7 is a plot of Component 1 against Component 2 from the loading matrix for states. Very high positive loadings were recorded for the state of Mississippi, New York, South Carolina and North Carolina in Component 1. Low loadings were recorded for Florida, Tennessee, Kentucky, Arizona, South Dakota and Alabama for the first Component in state loading matrix. Oklahoma, Alabama, North Dakota, and South Dakota had high positive loadings in Component 2 with Mississippi, Delaware and Maine having high negative loadings for the same component. Florida, Vermont, Wisconsin and Idaho also had low loadings in Component 2.

TABLE I. PARAFAC MODELS FITTED

Model	No. of Components	Var. Explained (%)	CORCONDIAG
1*	2	87.5311	67.353
2*	3	90.1485	34.3114
3*	4	91.8352	0.541
4	2	87.6218	92.0126
5	3	90.4965	45.1397
6	4	92.1787	5.2071

*-Nonnegative Model

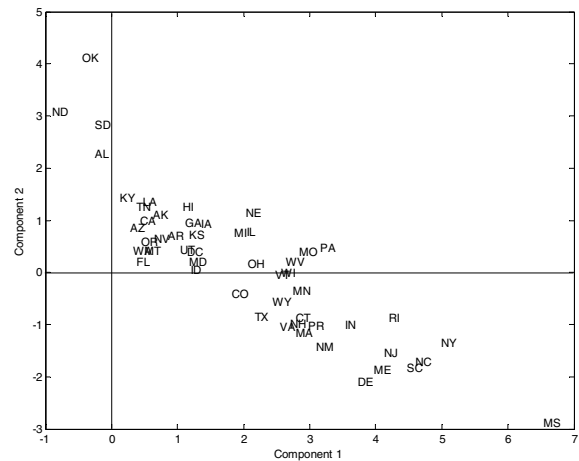


Fig. 7. Loadings Plot for US States

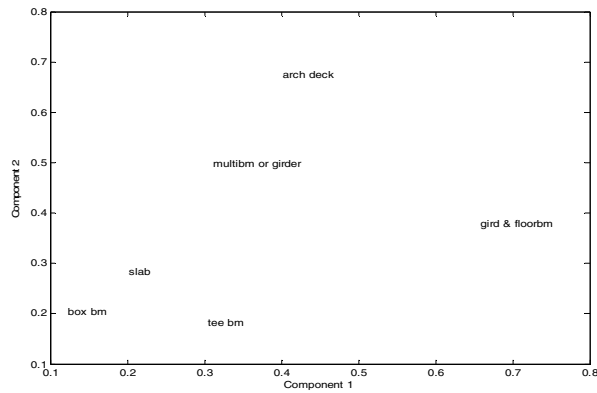


Fig. 8. Loading plot for bridge structure types

Loadings for structural type are shown in Fig. 8. Girder & floorbeams, arch-deck and multibeam or girder types had high loadings in the first component. Box Beam, slab and tee beams had low loadings in both components. However, girder and floorbeams had the highest loading in factor 1 while arch-deck had the highest loading in factor 2. Fig. 9 also shows the loading plot for time loading matrix. 2010, 2011, 2012, and 2013 all have low loadings on both components. In contrast, 1992, 1993, 1994, 1995 and 1996 all have high and moderate loadings for Component 1 and Component 2 respectively.

It can be deduced from the 3 loading plots that any increase along Component 1 implies a higher SD rate for girder & floorbeam systems. An increase along Component 2 indicates a higher SD rate for arch type and or stringer systems. Therefore, loading plot for states (Fig. 7) shows Mississippi, North and South Carolina having high loading scores along Component 1 implying high SD rates for girder & floorbeam systems in these states. However, they are located in the second quadrant which means they had relatively lower SD rates for arch deck and stringer systems. This is evident from the 3D visualization in Fig. 3. States such as Nebraska and Pennsylvania have relatively high SD rates for girder & floorbeam systems as well as arch deck and stringer systems and so can be located in the first quadrant. The loading plot for time also shows lower scores in Component 1 from 2010 to 2013. This is evidenced in the downward trend in the mean SD rates line plot in Fig. 6. 1992 to 1996 had high loading scores and this is because there were generally a high number of structurally deficient bridges

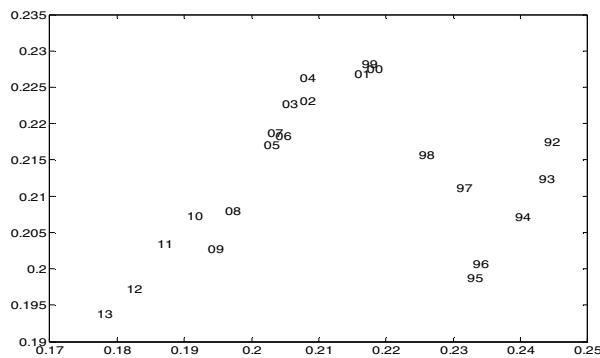


Fig. 9. Loading plot for time

nationwide. These plots demonstrate that the decomposition of tensor data results in factors that provide a simple and intuitive means to interpret multiway bridge data. The use of tensor decomposition can also help answer complex queries about the data under consideration.

Factors obtained after factorization can be used in clustering data. In the case of bridges, clustering may be important when similar trends and patterns are to be identified in a large multidimensional data set as used in this paper. Clusters obtained from analysis can be utilized as a guide for policy decisions as well as giving insights into design, environmental and other factors that can affect the performance of bridges. Clustering based on tensor decomposition factors provide satisfactory results. In this paper, bridge structural types were clustered based on the yearly structural deficiency ratings in all the states for a period of 22 years. Clusters obtained from averaging yearly SD ratings for each individual state yielded results that were slightly different from that of the tensor analysis. The averaging data clusters in Fig. 10 (i) shows tee beams as being closer to girder & floorbeam structure type in terms of behavior. Tensor factorization clusters (Fig. 10 (ii)) showed that the trend on the tee beam type was more similar to slab and box beams and that the multibeam and arch deck types were closer in behavior to the girder & floorbeam. It can be seen from Fig. 6 that the results from the tensor factorization clusters are much more in agreement with the actual trends within the analysis period where SD rates in tee beams are much closer to slab and box beams.

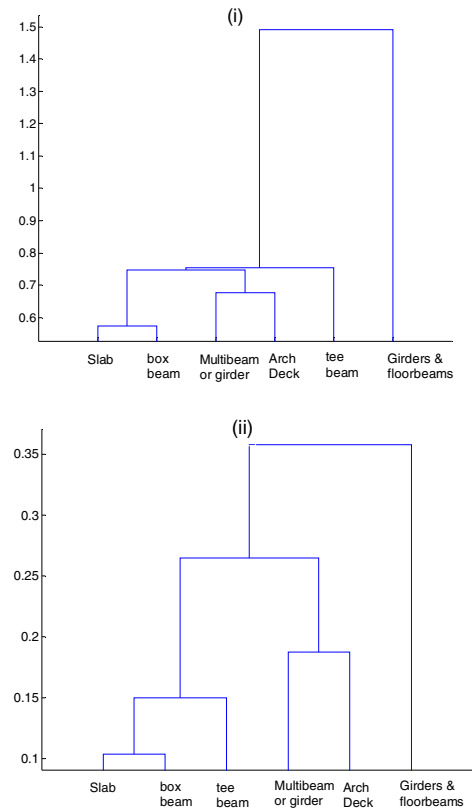


Fig. 10. Clusters of bridge structural types: (i) SD rate averaging clusters (ii) Tensor Factorization Clusters

V. CONCLUSION & RECOMMENDATIONS

This study demonstrates tensor decomposition as a promising new tool that will enhance knowledge discovery in bridge data analysis. Tensor decomposition was found to provide better insights into complex and multidimensional bridge data as opposed to the conventional 2-way analysis of data. In this study, it was found out that the girder & floorbeam structural types in bridges recorded the highest structural deficiency (SD) rates yearly. These were shown clearly from the loading plots which revealed girder & floorbeam systems as having high scores in Component 1. Higher scores along Component 2 indicated relatively high SD rates for Arch Deck and Multibeam or girder systems. The loading plots for the states showed how states varied with respect to SD rates of various structural types. Several clusters which make physical sense can be seen in the plot. Mississippi, with the highest SD rate for girder & floorbeam systems can be seen on the high side of component one while Florida, which has a very low SD rate for both girder & floorbeam and Arch deck types is located near the origin. The loading plot for the time mode shows a downward trend in recent years which is an accurate representation of the data.

It is recommended that a more detailed study of the NBI data including environmental factors as well as location information be incorporated into the data tensor. This multidimensional view of the bridge data will lead to results that will give a clearer representation of bridge asset condition. Also, since the loading matrices have factors which explain the variance in data, it is recommended that these will be utilized in predicting future conditions of bridge assets. Cluster analysis

based on tensor factorization can also be used in project prioritization and risk comparison by public agencies.

REFERENCES

- [1] C.A. Andersson and R. Bro, The N-way Toolbox for MATLAB, *Chemometrics & Intelligent Laboratory Systems*. 52(1):1-4, 2000.
- [2] Bridges. 2013 Report Card for America's Infrastructure. American Society of Civil Engineers.
<http://www.infrastructurereportcard.org/a/#p/bridges/overview>. Accessed: 8/23/2014
- [3] Wu N., Chase S. An Exploratory Data Analysis of National Bridge Inventory. *University of Virginia*. Mid-Atlantic Universities Transportation Center (MAUTC Regional III). May 2010
- [4] Structurally Deficient and Functionally Obsolete. Michigan Department of Transportation (MDOT). http://www.michigan.gov/mdot/0,4616,7-151-9618_47418-173622--,00.html. Accessed: 8/23/2014
- [5] Kolda T.G., Bader B.W. Tensor Decompositions and Applications*. *SIAM Review Vol 51*. No. 3, pp. 455-500. 2009
- [6] Singh K.P., Amrita M., Singh V.K., Sinha S. Multi-way data analysis of soils irrigated with wastewater- A case study. *Chemometrics and Intelligent Laboratory Systems* 83 (2006) 1-12
- [7] Morup M. Applications of tensor (multiway array) factorizations and decompositions in data mining. *WIREs Data Mining and Knowledge Discovery*. 2011 John Wiley and Sons, Inc. Volume 1 January/February 2011.
- [8] Dekelbab W. et al. History Lessons From the National Bridge Inventory. *U.S. DOT, Federal Highway Administration*. Publication Number: FHWA-HRT-08-004. Vol. 71, No. 6. May/June 2008.
- [9] A Guide to Important National Bridge Inventory Terms. National Bridges. *The National Bridge Inventory Database*. <http://nationalbridges.com/guide-to-ratings>. Accessed: 08/24/2014
- [10] Davis. S.L., Godlberg D. The Fix We're In For: The State of Our Nation's Bridges 2013. *Transportation for America*. Page 2. June 2013